

Python OOP Practice Problems

Concept apply → Enheritance, Polymorphism-Override,

Date-28-09-2025

Problem 1: Shape Management System

You are asked to create a simple program that manages shapes using Object-Oriented Programming in Python.

1. Create a base class called Shape with:
 - A method area() that returns 0.
 - A method perimeter() that returns 0.
2. Create two child classes that inherit from Shape:
 - Rectangle → takes width and height as inputs.
 - Circle → takes radius as input.
3. Override the methods area() and perimeter() in both child classes to calculate the correct values.
4. In the main program:
 - Create a list of shapes (Rectangle and Circle).
 - Use a loop to print the area and perimeter of each shape.

Problem 2: Banking System

You are designing a simple Banking System using Python OOP.

1. Create a base class called BankAccount with:
 - Attributes: account_number, balance
 - Methods:
 - deposit(amount)
 - withdraw(amount)
 - get_balance()
2. Create two child classes:
 - SavingsAccount → has an extra attribute interest_rate and a method add_interest().
 - CheckingAccount → has an extra attribute transaction_fee and overrides withdraw().
3. In the main program:
 - Create a list of accounts (some savings, some checking).
 - Deposit and withdraw money.
 - Apply interest to savings accounts.
 - Print the balances of all accounts.